

Hwayong Nam

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RESEARCH INTERESTS

Interests: Computer Architecture, Memory Systems, Near-data Processing, DRAM Microarchitecture, DRAM Reliability

EDUCATION

Ph.D candidate

Mar. 2021 – Present

Seoul National University

Prof. Jung Ho Ahn

B.S. in Electrical and Electronics Engineering

Mar. 2015 – Feb. 2021

Chung-Ang University

PUBLICATION

[**ASPLOS 2025**] "M5: Mastering Page Migration and Memory Management for CXL-based Tiered Memory Systems", Yan Sun, Jongyul Kim, Zeduo Yu, Jiyuan Zhang, Siyuan Chai, Michael Jaemin Kim, **Hwayong Nam**, Jaehyun Park, Eojin Na, Yifan Yuan, Ren Wang, Jung Ho Ahn, Tianyin Xu, Nam Sung Kim

[**ASPLOS 2025**] "Marionette: A RowHammer Attack via Row Coupling", Seungmin Baek, Minbok Wi, Seonyong Park, **Hwayong Nam**, Michael Jaemin Kim, Nam Sung Kim, Jung Ho Ahn

[**ISCA 2024**] "DRAMScope: Uncovering DRAM Microarchitecture and Characteristics by Issuing Memory Commands", **Hwayong Nam**, Seungmin Baek, Minbok Wi, Michael Jaemin Kim, Jaehyun Park, Chihun Song, Nam Sung Kim, Jung Ho Ahn

[**ICS 2024**] "CLAY: CXL-based Scalable NDP Architecture Accelerating Embedding Layers", Sungmin Yun, **Hwayong Nam**, Kwanhee Kyung, Jaehyun Park, Byeongho Kim, Yongsuk Kwon, Eojin Lee, Jung Ho Ahn

[**ASPLOS 2024**] "TAROT: A CXL SmartNIC-Based Defense Against Multi-bit Errors by Row-Hammer Attacks", Chihun Song, Michael Jaemin Kim, Tianchen Wang, Houxiang Ji, Jinghan Huang, Ipoom Jeong, Jaehyun Park, **Hwayong Nam**, Minbok Wi, Jung Ho Ahn, Nam Sung Kim

[**MICRO 2023**] "How to Kill the Second Bird with One ECC: The Pursuit of Row Hammer Resilient DRAM", Michael Jaemin Kim, Minbok Wi, Jaehyun Park, Seoyoung Ko, Jaeyoung Choi, **Hwayong Nam**, Nam Sung Kim, Jung Ho Ahn, Eojin Lee

[**CAL 2023**] "X-ray: Discovering dram internal structure and error characteristics by issuing memory commands", **Hwayong Nam**, Seungmin Baek, Minbok Wi, Michael Jaemin Kim, Jaehyun Park, Chihun Song, Nam Sung Kim, Jung Ho Ahn

[**TC 2023**] "GraNDe: Efficient Near-Data Processing Architecture for Graph Neural Networks", Sungmin Yun, **Hwayong Nam**, Jaehyun Park, Byeongho Kim, Jung Ho Ahn, Eojin Lee

[**CAL 2022**] "GraNDe: Near-data Processing architecture with adaptive matrix mapping for graph convolutional networks", Sungmin Yun, Byeongho Kim, Jaehyun Park, **Hwayong Nam**, Jung Ho Ahn, Eojin Lee

SKILLS

Skills: C/C++, Python, Verilog, SystemVerilog

TEACHING

Seoul National University, Teaching Assistant

Mar. 2021 - Aug. 2021

Programming Methodology